OASIS: Online Anchor-centric Scene Inference System for Open-Vocabulary Retrieval and Navigation in Dynamic Environments

Kaiyao Liao
Beihang University
Beijing, China
2406104@buaa.edu.cn

Qingfeng Li
Beihang University
Beijing, China
liqingfeng@buaa.edu.cn

Chen Chen
Hangzhou Innovation Institute of Beihang University
Hangzhou, China
chenx2ovo@buaa.edu.cn

Xiaoze Wu
Beihang University
Beijing, China
wwwww@buaa.edu.cn

Qing Sun
Hangzhou Innovation Institute of Beihang University
Hangzhou, China
hollysun@buaa.edu.cn

Jianwei Niu*
School of Computer Science and Engineering
Beihang University
Beijing, China
niujianwei@buaa.edu.cn

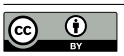
Abstract

Reliable robotic mapping in dynamic real-world environments is essential for embodied intelligence and long-term autonomy. However, existing methods often involve a trade-off: lightweight online mapping is efficient but captures limited semantic context, while dense mapping provides richer semantics at higher computational cost and is more vulnerable to distractors. They also struggle to relocalize objects after scene changes. To address these challenges, we propose the Online Anchor-centric Scene Inference System (OASIS), a unified system for online mapping, complex open-vocabulary retrieval, and navigation in dynamic environments. Motivated by the observation that indoor scenes are organized around stable semantic anchors, OASIS represents the environment using a sparse set of anchor nodes, each maintaining a unified multimodal representation that integrates semantic descriptions, visual features, and geometric states for reasoning, verification, and navigation. Its compact anchor-centric memory supports on-demand evidence extraction and hierarchical retrieval, reducing redundant storage while improving retrieval reliability. To handle scene dynamics, OASIS further reuses transient observations collected during navigation to relocalize moved objects. Experiments in both simulation and real-world environments, including standard retrieval benchmarks and navigation tasks under both static and dynamic settings, demonstrate that our system can perform complex open-vocabulary retrieval and navigation, while also enabling relocation of displaced objects after scene changes.

CCS Concepts

• **Computing methodologies** → **Scene understanding**.

*Corresponding author.



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Semantic Scene Graph, Open-Vocabulary Retrieval, Robotic Navigation

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1 Introduction

Modern embodied agents face increasingly complex tasks in unstructured environments, where instructions involve rich, open-vocabulary descriptions beyond simple object categories. A user might command a robot to "find the sealed silver can on a round table" or "locate the bed with a lit lamp." Fulfilling such requests requires a scene representation that transcends traditional geometric mapping, demanding fine-grained attribute recognition (*e.g.*, color, material, state) and spatial relationship reasoning. Crucially, these capabilities must remain robust in dynamic real-world settings where objects are frequently repositioned. Consequently, the challenge has shifted from static mapping to constructing a dynamic, semantic-rich scene graph capable of supporting intricate queries and adapting to environmental changes on the fly.

However, deploying semantic maps in real-world systems still involves a practical trade-off between efficiency and reasoning capability. Dense mapping approaches [6, 18, 28] embed rich semantics into many spatial elements, providing detailed understanding but sometimes introducing clutter and distractors, while also requiring relatively high computational and memory resources. Conversely, lightweight methods [9, 15] are more efficient, but may not preserve enough spatial context for complex reasoning, making it harder to distinguish visually similar objects based on their surroundings or attributes. A remaining challenge appears in dynamic environments. Although some methods address object relocalization, they often depend on repeated re-observation. During active navigation, rapid

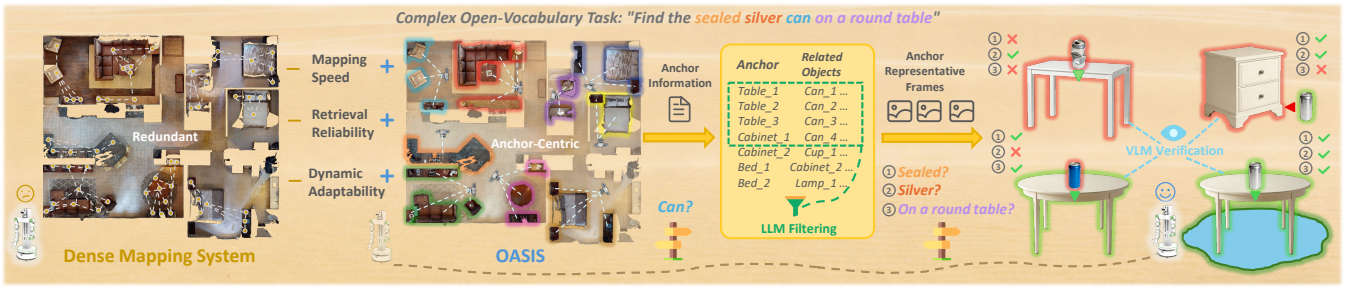


Figure 1: Comparison of retrieval paradigms for complex open-vocabulary queries. Unlike traditional dense mapping (left) which struggles with redundant context and visual distractors, our OASIS (right) constructs a lightweight anchor-centric memory. By leveraging an LLM to filter irrelevant regions and performing on-demand VLM verification, OASIS efficiently isolates the target based on fine-grained attributes and spatial relationships, significantly improving retrieval reliability.

robot motion and a limited camera view can cause shifted targets to be missed, which reduces the efficiency of these scanning-based strategies.

To bridge this gap, we propose the Online Anchor-centric Scene Inference System (OASIS) for robust open-vocabulary retrieval and dynamic object relocalization. As illustrated in Figure 1, OASIS avoids processing redundant context globally through an on-demand retrieval strategy. Our key insight is that indoor environments are inherently structured around stable, large-scale objects (*e.g.*, beds, tables, sofas), which we define as semantic anchors. Inspired by human episodic memory, OASIS constructs an anchor-centric graph rather than a dense per-pixel map, retaining only salient semantic descriptions and high-quality representative frames. During retrieval, a Large Language Model (LLM) first reasons over the semantic anchors to filter irrelevant regions, followed by on-demand Vision-Language Model (VLM) verification of the retrieved representative frames. This strategy keeps memory lightweight while preserving the ability to distinguish fine-grained visual attributes. In dynamic settings, OASIS additionally maintains transient observations along the navigation trajectory in a lightweight buffer; if a target is missing at its initial anchor, the system queries this buffer and re-verifies candidate views for reliable relocalization.

Our main contributions are summarized as follows:

- We present OASIS, an **online anchor-centric scene inference system** that unifies online mapping, open-vocabulary retrieval, and navigation in dynamic environments.
- We propose a human-inspired **episodic memory curation mechanism** that uses scale-aware scoring and spatio-semantic clustering to select representative frames, yielding a multimodal **anchor-centric scene graph** with integrated semantic, visual, and geometric representations.
- Built upon this scene graph, OASIS supports **hierarchical coarse-to-fine retrieval** and view-driven navigation, and introduces a **Transient Visual Buffer** to reliably relocalize moved objects after scene changes.
- We conduct extensive simulation and real-world experiments on retrieval and navigation under static and dynamic settings, demonstrating strong performance and reliable object relocation after scene changes.

2 Related Work

2.1 Open-Vocabulary 3D Scene Understanding

Closed-set semantic mapping relies on detectors trained for fixed object categories [8, 16, 17, 24]. Open-vocabulary methods instead distill features from vision-language models (*e.g.*, CLIP [20]) into 3D representations. Dense mapping methods [6, 18, 28] project pixel-aligned features into 3D space, enabling fine-grained queries but incurring high computational and memory costs. Conversely, object-centric or lightweight semantic point cloud methods [9, 15] construct sparse maps by aggregating features at the instance level. While these approaches are significantly more efficient, they often discard the surrounding visual context required for complex reasoning. Our work builds upon the efficiency of lightweight representations but augments them with an implicit storage mechanism to retain rich contextual details without the overhead of dense fields.

2.2 Retrieval and Reasoning on Scene Graphs

Recent advancements in open-vocabulary scene understanding have focused on reasoning about and locating objects within complex environments using foundation models. Early approaches primarily utilized CLIP-based matching [6, 19, 26, 27], which directly computes cosine similarity between text queries and stored visual features. While efficient for simple identification, these methods lack the capability for multi-step reasoning or handling spatial relationships. To overcome these limitations, LLM-based methods [7, 13, 23, 28, 31] leverage the reasoning power of LLMs to infer potential locations or generate navigation plans based on textual abstractions of the scene. However, relying solely on text descriptions often results in the loss of critical visual details (*e.g.*, object states or specific textures) necessary for precise identification. More recently, VLM-based approaches [18, 35] have emerged, directly processing images to retain visual fidelity. Despite their superior understanding, these methods typically suffer from high latency and computational costs, making them less practical for real-time applications. Our work bridges these gaps by introducing an efficient VLM-driven mechanism that operates on implicit storage, enabling detailed visual reasoning without the heavy overhead of full-image processing.

2.3 Object Navigation

Object navigation requires agents to locate and move towards specific target instances within an environment [2, 3, 5, 12, 14, 21, 30, 32–34]. Traditional approaches typically rely on static semantic maps, assuming that objects remain fixed throughout the mission. These methods combine SLAM with semantic segmentation to build a persistent representation, allowing the agent to plan paths to previously observed locations. However, real-world environments are inherently dynamic. Existing works on dynamic navigation [9, 27] typically address this by relying on continuous fine-grained modeling and geometric matching of objects along the trajectory to update the map. This strategy faces significant challenges: navigation often involves transient observations, making it difficult to capture and update the state of small objects accurately in real-time. Furthermore, when a target is missing from its expected location, these methods often lack the active reasoning capability to deduce its whereabouts. In contrast, our proposed method leverages subtle visual details observed along the path, enabling the system to infer the new location of displaced objects based on semantic context, even if the new location lies outside the immediate navigation trajectory.

3 Method

3.1 Overview

As illustrated in Figure 2, OASIS addresses the challenge of simultaneous online mapping and open-vocabulary retrieval in dynamic environments. From the RGB-D stream and robot localization, it first performs online semantic mapping [9] to maintain an instance set, identify stable anchors, and associate observations with anchors; in parallel, it curates a compact episodic memory through spatial-visual clustering and scale-aware scoring that adaptively balances local fidelity and global richness to keep only high-value representative frames; finally, it fuses anchors, representative frames, structural relations, and navigable poses into the OASIS graph for coarse-to-fine candidate retrieval, VLM-based verification, and navigation, while a relocation branch handles target relocalization when objects are moved. For full reproducibility, exhaustive implementation details—including complete LLM/VLM prompts, exact mathematical formulations, and hyperparameter settings—are provided in the Supplementary Material.

3.2 Episodic Memory Curation

Inspired by the human cognitive process of episodic memory, which retains only significant events and high-quality visual snapshots rather than continuous redundant streams, we design a dynamic memory curation mechanism. This module processes the input frame sequence $\mathcal{F} = \{f_1, f_2, \dots\}$ to maintain a set of spatial-visual clusters $\mathcal{C} = \{C_1, \dots, C_N\}$, where each cluster C_k is represented by a single optimal frame r_k.

LLM-Driven Anchor Definition. Prior to memory curation, we explicitly define the set of target objects—termed *Anchor Objects*—to align with human spatial cognition. Instead of manual selection, we leverage an LLM to identify object categories that satisfy three key criteria: (1) Volumetric Prominence (occupying significant space), (2) Stationarity (structurally static or heavy), and (3) Associative Utility (serving as natural landmarks for referencing nearby items). This LLM-driven approach ensures our system can generalize across diverse indoor environments by focusing on semantically stable landmarks (*e.g.*, *bed*, *sofa*) while filtering out less informative observations to keep the representation focused.

Scale-Aware Frame Scoring. To retain only informative observations, we score each incoming frame by combining global scene richness and local anchor quality:

$$S(f_i) = w_{\text{global}} S_{\text{global}}(f_i) + w_{\text{local}} S_{\text{local}}(f_i), \tag{1}$$

$$S_{\text{global}}(f_i) \propto \alpha \min(M, A_{\max}) + (1-\alpha)\min(N_{\text{obj}}, O_{\max}), \tag{2}$$

$$S_{\text{local}}(f_i) \propto \gamma \max_j(V_j) + (1-\gamma)\frac{1}{M}\sum_j V_j, \tag{3}$$

$$V_j = \beta E_j + (1-\beta)\min(C_j, C_{\max}), \tag{4}$$

Here, M and N_{obj} denote the numbers of visible anchors and non-anchor objects in frame f_i, respectively. $A_{\max}$, $O_{\max}$, and $C_{\max}$ are clipping bounds for anchor count, object count, and interaction count. For each anchor j, $E_j \in [0,1]$ denotes edge completeness and C_j denotes interaction count. The coefficients $\alpha, \beta, \gamma \in [0,1]$ balance the corresponding terms. Both S_{global} and S_{local} are normalized to $[0,1]$; when no anchor is visible ($M = 0$), we set $S_{\text{local}} = 0$.

The adaptive weights are determined by anchor coverage and anchor-area balance. Let a_j denote the 2D image area of anchor j in frame f_i, and define the normalized area distribution as $p_j = a_j / \sum_m a_m$, with entropy $H(f_i) = -\sum_j p_j \log p_j$, and the anchor coverage ratio as $r(f_i) = \sum_j a_j / A_{\text{frame}} \in [0,1]$, where A_{frame} is the image area. We then compute

$$w_{\text{global}} = \begin{cases} 1 - r(f_i), & M \le 1, \\ (1-\lambda)(1-r(f_i)) + \lambda \dfrac{H(f_i)}{\log M}, & M > 1, \end{cases} \tag{5}$$

and set $w_{\text{local}} = 1 - w_{\text{global}}$, where $\lambda \in [0,1]$ balances coverage and entropy. Intuitively, frames dominated by a few large anchors emphasize local fidelity, while frames with smaller or more balanced anchors emphasize global context.

Cluster Assignment and Representative Frame Update. Armed with the frame score $S(f_i)$, we aim to curate a compact yet expressive memory from the highly redundant continuous RGB-D stream. We maintain the memory as a set of spatial-visual clusters. Each cluster C_k is characterized by an aggregated visual feature v_{C_k} and a *representative frame* r_k—the highest-scoring frame that best encapsulates the visual and semantic information of that cluster's viewpoint.

To process the incoming stream, we first assign f_i to an existing cluster. We extract its visual feature vector v_i using a pre-trained CLIP encoder and obtain the camera pose p_i from the localization module. A frame f_i is assigned to C_k if it satisfies both visual and spatial proximity:

$$\text{Sim}_{\text{vis}}(v_i, v_{C_k}) > \tau_{\text{vis}} \quad \text{and} \quad \text{Sim}_{\text{spa}}(p_i, p_{r_k}) > \tau_{\text{spa}}, \tag{6}$$

where Sim_{vis} denotes the cosine similarity, and p_{r_k} is the pose of the cluster's representative frame r_k.

If f_i matches C_k, the cluster is updated in two ways. First, the cluster feature v_{C_k} is dynamically updated via a weighted fusion parameterized by the frame scores $S(\cdot)$, ensuring that higher-quality frames contribute more significantly to the cluster representation.

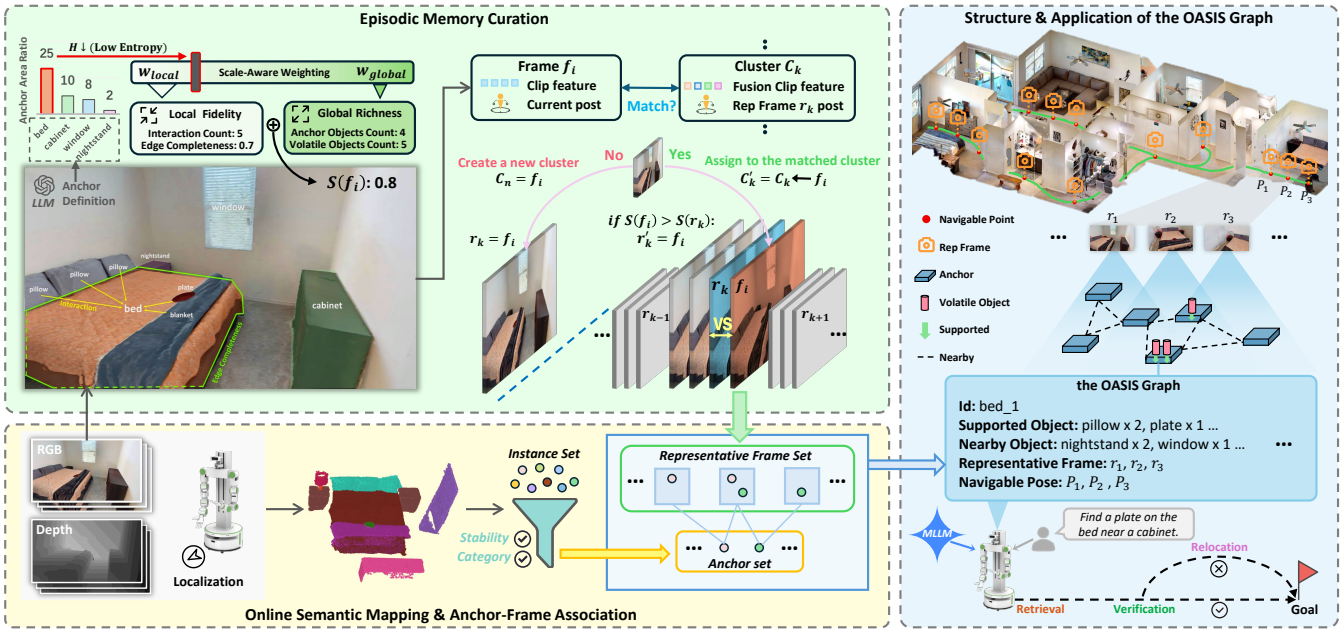


Figure 2: System architecture of OASIS. Given an RGB-D stream, the system concurrently performs episodic memory curation (top left) and online semantic mapping (bottom left) to extract 3D instances. To build the episodic memory, it utilizes scale-aware scoring and spatial-visual cluster assignment to continuously update and retain high-quality representative frames. By assigning these selected frames to their corresponding stable anchor instances, the system constructs the unified multimodal OASIS Graph, structured as shown on the right. Leveraging this graph alongside multimodal foundation models (LLM/VLM), the agent effectively executes hierarchical retrieval and robust view-driven navigation in dynamic environments.

Second, we update the representative frame r_k strictly when the new frame offers superior information density:

$$r_k \leftarrow f_i \quad \text{if} \quad S(f_i) > S(r_k). \tag{7}$$

Conversely, if f_i presents novel visual content or a distinct viewpoint (failing Equation 6), it initializes a new cluster C_n with itself as the initial representative frame ($r_n = f_i$), thereby progressively expanding the system's spatial-visual coverage. In our implementation, we set $\tau_{\text{vis}} = 0.8$, and define the spatial proximity τ_{spa} using translation and rotation bounds of 2 m and 45°, respectively.

3.3 Anchor-Centric Scene Graph Construction

Building upon episodic memory, we construct the Anchor-Centric Scene Graph (the OASIS graph). This process integrates dense 3D information from the geometric mapping stream with high-quality visual snapshots from the memory stream into a unified multimodal representation.

Online Semantic Mapping and Anchor-Frame Association. Concurrently with memory curation, we leverage an online semantic mapping process [9]. Let $o_i \in O$ denote an observed instance. The mapping process retains stable instances $o_i^{\text{st}} \in O^{\text{st}}$ that persist for more than n consecutive frames, and provides instance-level 3D boxes, point-cloud support, CLIP features, and high-confidence vertical relations (*supported*) from box overlap. We then use the anchor categories defined in Sec. 3.2 to obtain anchor instances A_j from O^{st}, and assign representative frames by intersecting each anchor's visible-frame set with memory frames $\mathcal{R}$. For compactness, if an anchor A_j is linked to more than K candidates, we keep the top-K frames (default $K = 5$) ranked by $S(\cdot)$.

Graph Formalization. Formally, we define the OASIS graph as a graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}\}$, where $\mathcal{V}$ denotes the set of stable anchor instances. Unlike traditional scene graphs that attempt to exhaustively classify all relationships into static text labels, our OASIS graph functions as a multimodal index, prioritizing high-confidence geometric links while preserving rich visual contexts. The edges associated with an anchor A_i are defined as:

- **Geometric Edges:** We define directed edges $O_j \xrightarrow{\text{on}} A_i$ for objects physically supported by anchor A_i, derived from 3D bounding-box analysis.
- **Visual Edges:** Each anchor connects to its representative frames $\mathcal{R}_i = \{r_{i,1}, \ldots, r_{i,K}\}$; stable instances that co-occur in these frames may exhibit diverse relative layouts (e.g., left, right, front, or behind), which we abstractly define as *nearby* and use to form the anchor's *nearby-list* for subsequent retrieval and verification.

In our implementation, the graph is serialized in JSON format. To maintain a lightweight structure, the visual edges store the file paths of the representative frames rather than raw image data. Additionally, each frame entry records its corresponding 6-DoF camera pose, which is utilized in subsequent navigation steps.

3.4 Hierarchical Object Retrieval

Leveraging the structured OASIS graph, we propose a three-stage *coarse-to-fine* retrieval algorithm. This approach avoids the computational cost of scanning the entire scene by prioritizing semantically relevant anchors before performing detailed visual verification.

Stage 1: Semantic Anchor Proposal (LLM-based). Given a user query Q (*e.g.*, "*Find the mug near the laptop*"), the system first reasons about the most probable locations using only the textual layer of the OASIS graph. An LLM functions as a planner, taking the query and the global anchor list as input. For each anchor A_i, the LLM analyzes its semantic category and the summary of its related objects (both *supported-list* and *nearby-list*). The LLM outputs a ranked list of candidate anchors $\mathcal{A}_{cand}$ (Top-3 by default) where the target is most likely to appear. This step purely relies on textual reasoning, effectively pruning the search space without processing any images.

Stage 2: Visual Verification with Cross-Anchor Reasoning (VLM-based). The system iterates through candidate anchors in $\mathcal{A}_{cand}$. For each queried anchor A_i, it retrieves representative frames and projects 2D boxes with IDs for *all visible anchors* (not only A_i), enabling cross-anchor disambiguation in shared views where multiple anchors co-occur. The VLM then verifies target presence for query Q, identifies the associated anchor $A_{selected}$ (which may differ from A_i), and outputs the selected anchor ID, the target relation to $A_{selected}$ (*i.e.*, `self`, `on`, or `nearby`), and a detailed visual description with predicted class for subsequent instance matching.

Stage 3: Precise Instance Localization (CLIP-based). Once the VLM confirms a detection and identifies the reference anchor $A_{selected}$, we map the high-level perception back to a specific 3D instance UID (u_{target}). When the returned relation is `self`, the UID of $A_{selected}$ is directly returned. Otherwise, for relations `on` or `nearby`, we retrieve the corresponding candidate list (the *supported-list* or *nearby-list*) from node $A_{selected}$ and first filter candidates by the VLM-predicted class. When class-matched candidates exist, ambiguity (*e.g.*, multiple mugs) is resolved by cosine similarity between candidate CLIP embeddings and the text embedding of the VLM-generated *Visual Description*; when no class match exists (*e.g.*, open-vocabulary mismatch), we compute CLIP similarity over the full candidate list and select the highest-scoring instance as the final target.

Global Fallback Mechanism. In cases where the hierarchical search fails (*i.e.*, the target is not found in any candidate anchor's view), the system triggers a global fallback. We perform a global CLIP-based search across all scene instances using the visual description generated during the failure analysis (or the original query). To ensure efficiency, instances already examined during Stage 2 are excluded.

3.5 Navigation and Relocation

To close the perception-action loop, our system employs a view-driven navigation strategy integrated with a two-tier relocation mechanism, ensuring robustness against dynamic environmental changes.

View-Driven Navigation & Verification. Upon retrieving the target instance u_{target} in Sec. 3.4, the system determines the optimal observation pose. For targets identified via VLM (Stage 2), the frame explicitly marked as the *best view* is selected; for those retrieved via CLIP fallback (Stage 3), the frame with the highest similarity score among all stored observations is chosen. The robot then navigates to the recorded 6-DoF pose T_{wc} of the selected frame, aligning its camera to replicate the original viewpoint. Upon arrival, the robot captures a current observation I_{curr}. The VLM performs a visual comparison between the retrieved memory frame I_{mem} and I_{curr} to confirm the target's presence.

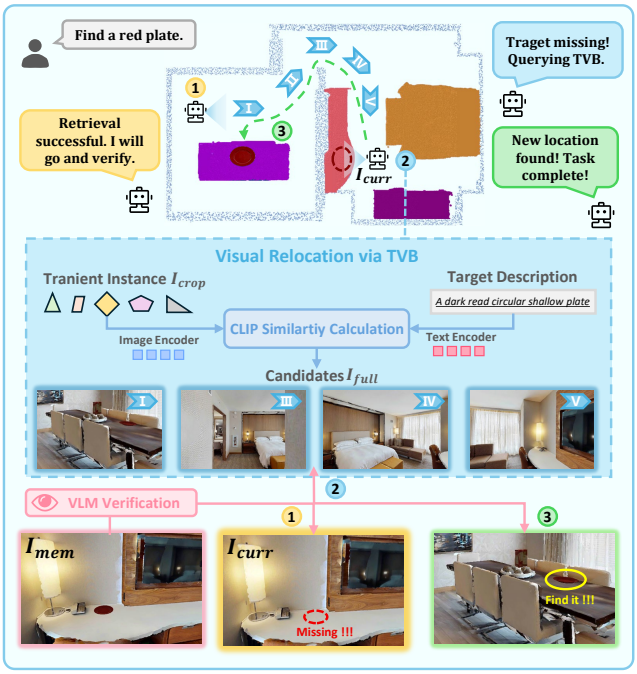


Figure 3: Visual relocalization via Transient Visual Buffer (TVB). If the retrieved target is absent from the current view, OASIS uses CLIP to match transient crops to the query and a VLM to verify candidate frames and confirm the target's new location.

Dynamic Object Relocation. Should the verification fail (*i.e.*, the object has moved), the system triggers the relocation procedure illustrated in Figure 3. Since the robot does not perform dense mapping during navigation, we introduce a lightweight **Transient Visual Buffer** to capture potential sightings of the moved object.

1) Transient Instance Capture: During navigation, we reuse the same object detection pipeline adopted in the mapping stage to process the video stream in real time. The matching step is performed against objects already stored in the semantic map; detections that cannot be matched to any mapped instance are flagged as "Transient Instances". For each such instance, we cache a tuple $\langle I_{crop}, I_{full}, T_{pose} \rangle$, storing the cropped object image, the full contextual frame, and the camera pose. To conserve memory, only the highest-confidence frame is retained for each transient track.

2) Visual Relocation: When the target is reported missing, the system first queries this buffer. We compute CLIP similarities between the target's visual description (from Sec. 3.4) and the cached

Table 1: Fine-grained object retrieval. Comparison on the OpenLex3D (Replica) benchmark.

Method	mAP ↑	AP_{50} ↑	AP_{25} ↑
ConceptGraphs [6]	2.74	6.18	13.82
HOV-SG [28]	3.72	7.59	14.94
Kassab2024 [10]	0.66	2.15	5.34
OpenMask3D [26]	2.10	4.93	12.63
DualMap [9]	6.19	11.72	20.06
Ours	**6.58**	**12.52**	**20.63**

I_{crop} images, selecting the Top-5 candidates. These 5 full frames, along with the original target image I_{target}, are fed into the VLM. The VLM is prompted to determine if the missing target appears in any of the new locations. Upon a positive match, the robot updates the target's position and navigates to the corresponding T_{pose} to perform the final verification as described above.

3) Semantic Relocation: In scenarios where the visual search yields no matches (implying the object was not seen along the path), the system reverts to semantic reasoning. The LLM analyzes the OASIS graph again, excluding the current location, to predict the next most probable anchor (*e.g.*, "*If the mug is not on the desk, check the coffee table nearby*"). The robot then navigates to the pose of the representative frame with the highest $S(\cdot)$ score for this predicted anchor and repeats the verification process.

4 Experiments

4.1 Overview

In our implementation, we use GPT-5.2 for LLM planning and Gemini 3 Pro for VLM verification. Since our main contribution is to improve semantic understanding and dynamic relocation capabilities, our evaluation across diverse retrieval and navigation tasks focuses on answering three key questions: (1) **Complex Retrieval Capability:** Can OASIS accurately retrieve objects based on complex language queries? (2) **Dynamic Adaptability:** Can the system effectively relocate objects that have moved after the initial mapping phase? (3) **Component Efficacy:** What is the contribution of each module to the overall performance? Crucially, OASIS preserves the real-time efficiency required for online operation; detailed system profiling (*e.g.*, mapping speed and inference latency) is provided in the Supplementary Material.

4.2 Object Retrieval

4.2.1 Experimental Setup. To comprehensively evaluate OASIS, we conduct experiments on three benchmarks spanning Replica [25], HM3D [22], and ScanNet [4] scenes, each targeting a different retrieval capability.

1. Fine-grained Object Retrieval. We utilize the OpenLex3D benchmark [11] (Replica subset) to assess fine-grained object-level retrieval performance. Its human-annotated queries cover all scene objects and combine pure-synonym and synonym+descriptor forms, including fine-grained pattern, cultural, and brand cues (*e.g.*, "floral cup", "ironman portrait", "nike sneaker"), making the retrieval task more realistic and challenging.

Table 2: Functional region retrieval. Comparison on the HOV-SG (HM3D) benchmark.

Method	Regions	Trials	SR (%) ↑
ConceptGraphs [6]	15.6	34.9	29.3
HOV-SG [28]			31.5
DualMap [9]			53.8
Ours			**57.7**

Metric: Following OpenLex3D, we report mean Average Precision (mAP), as well as AP at IoU thresholds of 0.25 (AP_{25}) and 0.50 (AP_{50}).

2. Functional Region Retrieval. To evaluate macro-level scene understanding, we follow the HOV-SG room retrieval protocol [28] and its annotated room–object ground truth on eight HM3D scenes (15.6 regions/scene on average). This setup tests whether the model can accurately localize broad functional areas (*e.g.*, "kitchen", "office") rather than isolated objects, under consistent task definitions and supervision.

Metric: Following HOV-SG, we report Top-5 accuracy over 20 frequent object categories; a query is successful if the predicted object overlaps the ground truth with IoU > 0.1.

3. Spatial Relation Retrieval. The goal is to evaluate robust spatial relation retrieval under varying contextual complexities and viewpoint dependencies. We use the Sr3D+ benchmark [1], built on ScanNet scenes, and report four standard splits: *Easy* (≤2 distractors; *e.g.*, "The lamp closer to the white armchair."), *Hard* (>2 distractors; *e.g.*, "The lamp on the right in between the beds."), *View-Dependent* (needs a specific view; *e.g.*, "Looking at bed, the one on the left."), and *View-Independent* (viewpoint-free; *e.g.*, "The large cabinets directly above the fridge and oven.").

Metric: Following CORE-3D [18], we report grounding accuracy at IoU thresholds of 0.1 and 0.25 (A@0.1, A@0.25).

4.2.2 Results and Analysis. As presented in Table 1, our method achieves the best performance in **Fine-grained Object Retrieval** (6.58 mAP). The margin over DualMap is modest because both methods share the same instance-level scene construction pipeline. Our gain mainly comes from finer-grained attribute recognition, which yields higher feature similarity for the true target and pushes it to earlier ranks; this improves per-query AP and, consequently, leads to a slightly higher overall mAP. In **Functional Region Retrieval** (Table 2), our method effectively handles this task. Although our scene graph has no explicit room-level hierarchy, our system determines functional areas by semantic reasoning over objects around each anchor together with visual verification of background layout. Table 3 details the **Spatial Relation Retrieval** results. Our method excels in the "Easy" and "View-Independent" splits, outperforming the specialized baseline CORE-3D. This confirms that our LLM planner effectively resolves local spatial constraints (*e.g.*, "next to"). In the "Hard" split, CORE-3D remains stronger, which suggests a natural boundary of our current design. These hard queries often involve long-range relationships (*e.g.*, "farthest from") that span multiple anchors. Because each anchor representative frame usually captures objects near that anchor, our anchor-centric verification can be less effective when the required spatial relation is between far-apart objects.

Table 3: Spatial relation retrieval. Grounding accuracy on the Sr3D+ (ScanNet) benchmark across different difficulty splits.

Method	Overall (%)		Easy (%)		Hard (%)		View-Dependent (%)		View-Independent (%)	
	A@0.1 ↑	A@0.25 ↑	A@0.1 ↑	A@0.25 ↑	A@0.1 ↑	A@0.25 ↑	A@0.1 ↑	A@0.25 ↑	A@0.1 ↑	A@0.25 ↑
OpenFusion [29]	12.6	2.4	14.0	2.4	1.3	1.1	3.8	2.5	13.7	2.4
BBQ-CLIP [13]	14.4	8.8	15.4	9.0	6.7	6.7	11.4	5.1	14.4	8.8
ConceptGraphs [6]	13.3	6.2	13.0	6.8	16.0	1.3	15.2	5.1	13.1	6.4
DualMap [9]	28.4	21.6	29.9	22.4	17.3	16.0	16.5	11.4	30.1	23.0
BBQ [13]	34.2	22.7	34.3	22.7	33.3	22.7	**32.9**	20.3	34.4	23.0
CORE-3D [18]	41.8	35.6	41.8	35.7	**41.3**	**34.7**	**32.9**	**30.4**	43.0	36.3
Ours	**44.0**	**37.4**	**44.7**	**37.9**	38.7	33.3	**32.9**	29.1	**45.5**	**38.5**

Table 4: Object Navigation Success Rate (SR) on HM3D.

Scene Type	Method	00829	00848	00880	Trials	Avg. SR ↑
Static	ConceptGraphs [6]	69.2	53.8	61.5	78	61.5
	HOV-SG [28]	53.8	46.2	57.7		52.6
	DualMap [9]	73.1	69.2	69.2		70.5
	Ours	**76.9**	**92.6**	**84.6**		**85.9**
In-Anchor	DualMap [9]	66.7	66.7	61.1	54	64.8
	Ours	**83.3**	**83.3**	**77.8**		**81.5**
Cross-Anchor	DualMap [9]	55.6	61.1	64.7	53	60.3
	Ours	**77.8**	**83.3**	**70.6**		**77.4**

4.3 Navigation and Relocation

4.3.1 Experimental Setup. We evaluate closed-loop navigation performance on the HM3D dataset [22], adhering to the protocol established in DualMap [9]. The evaluation encompasses three scenarios: **Static**, where the environment remains unchanged; **In-Anchor Dynamic**, where target objects are moved within the same anchor region (*e.g.*, shifted on a table); and **Cross-Anchor Dynamic**, where objects are relocated to entirely different rooms or distant anchors. We report Success Rate (SR), defined as the percentage of queries in which the agent stops within 1 meter of the queried object. In dynamic scenes, a query is counted as successful only if the target is found within three attempts.

4.3.2 Results and Analysis. Table 4 summarizes the navigation results, where our method consistently outperforms all baselines. In the **Static** setting, our method achieves an impressive average Success Rate (SR) of 85.9%, significantly surpassing DualMap (70.5%) and ConceptGraphs (61.5%). This substantial performance gain stems primarily from the superior retrieval accuracy of OASIS. Our approach also shows clear advantages in dynamic scenarios. For **In-Anchor** moves, our method maintains a high SR of 81.5%, demonstrating robustness to local object displacements. The most meaningful performance gain, however, is observed in the challenging **Cross-Anchor** setting, where our method achieves 77.4% SR compared to DualMap's 60.3%. DualMap adopts a strategy of continuous scene graph updating during navigation, which generally requires intentional scanning to capture small objects along the trajectory. By contrast, our Transient Visual Buffer (Sec. 3.5) passively captures transient object observations during robot transit, enabling efficient visual relocalization of displaced targets. Even if no new target location is detected along the navigation path, our semantic relocalization mechanism can further infer potential positions to support subsequent verification.

Table 5: Memory curation ablation. Impact of different clustering and frame selection strategies on scene instance coverage.

Clustering	Global	Local	Adaptive	Avg. Clusters ↓
Spatial-Only	87.18	83.08	83.99	5.3
Visual-Only	84.79	72.92	86.64	**3.8**
Hybrid (Ours)	**90.82**	**89.30**	**91.13**	6.5

4.4 Ablation Studies

4.4.1 Memory Curation Strategy. We analyze the interplay between *Clustering* and *Frame Selection* using a full factorial design that combines three clustering methods (Spatial-Only, Visual-Only, Hybrid) with three selection scores (Global, Local, Adaptive Fusion). We report Instance Coverage, defined as the fraction of ground-truth Replica instances visible in at least one representative frame, and the average cluster count; all results are averaged over eight scenes.

As shown in Table 5, *Visual-Only* clustering yields the fewest clusters but lower coverage because global CLIP features neglect spatial structure, causing confusion between distinct yet visually similar frames and different viewpoints. *Spatial-Only* provides moderate coverage but cannot distinguish semantically different frames in close proximity. In contrast, *Hybrid* achieves the highest coverage (>89%) across all selection methods with a moderate cluster count. For frame selection, *Local* underperforms and *Global* remains stable, whereas *Adaptive* reaches 91.13% coverage, confirming the benefit of combining global context with local quality.

4.4.2 Hierarchical Retrieval Pipeline. We evaluate the contribution of each coarse-to-fine stage on Sr3D+ using three variants: (1) *Full ablation* removes hierarchical reasoning and selects the highest-CLIP instance across the scene; (2) *w/o LLM Proposal* obtains top-ranked instances through global CLIP matching, maps them to three unique anchors while skipping duplicates, and verifies their representative frames with the VLM; and (3) *w/o VLM Verification* retains LLM-based anchor proposals but uses CLIP for final matching.

Table 6 confirms the necessity of both stages. Full ablation performs worst without spatial reasoning or fine-grained validation.

Table 6: Hierarchical retrieval ablation. Impact of LLM proposal and VLM verification stages on the Sr3D+ benchmark.

LLM reasoning	VLM verification	A@0.1 (%) ↑	A@0.25 (%) ↑
✗	✗	28.4	21.6
✗	✓	36.5	30.2
✓	✗	32.8	26.9
✓	✓	**44.0**	**37.4**

Table 7: Dynamic navigation ablation. Impact of system components on relocalization performance in the Cross-Anchor setting.

Method	Attempts ↓	SR (%) ↑	SPL (%) ↑
Random Anchor	2.9	17.0	3.9
Ours w/o Transient Visual Buffer	2.5	56.6	17.3
Ours w/o Semantic Reasoning	1.8	73.6	24.1
Ours (Full)	**1.6**	**77.4**	**25.8**

The *w/o LLM Proposal* variant retains reasonable accuracy but incurs high latency from global matching and unpruned anchor verification. The *w/o VLM Verification* variant loses precision because CLIP cannot resolve fine-grained visual ambiguity or exact spatial relations. The full pipeline achieves the best accuracy–efficiency trade-off.

4.4.3 Dynamic Navigation and Relocation. Finally, we assess navigation robustness in the *Cross-Anchor* setting where objects are relocated to distant regions. We report Attempts (the number of relocation attempts per query) and SPL (success weighted by path length). We compare our full framework against a *Random Anchor* baseline and two ablated variants: the baseline randomly navigates to the pose of the best representative frame of a random anchor, while the variant without Transient Visual Buffer performs semantic relocation at every step, and the one without Semantic Reasoning reverts to random anchor navigation if no new target location is detected in the buffer.

Quantitative results in Table 7 demonstrate the superiority of our complementary design. The *Random Anchor* baseline performs poorly with 17.0% SR and 2.9 average attempts due to blind random search in the large scene space. Removing the visual buffer leads to limited performance (56.6% SR) as pure semantic inference lacks reliable visual cues. Ablating semantic reasoning achieves moderate results (73.6% SR) but suffers from inefficient random fallback. In contrast, our full framework integrates both components and achieves the best performance of 77.4% SR with only 1.6 average attempts, validating the effectiveness of our dynamic relocalization mechanism.

4.5 Real-World Experiments

As shown in Figure 4, we construct a real-world household environment comprising a bedroom, kitchen, and living room, with 13 semantic anchors. The scene includes visually similar objects with diverse categories, colors, shapes, and states, providing realistic distractors for open-vocabulary retrieval. We then conduct 30 retrieval-and-navigation trials in each of the static and dynamic settings. Static trials use fine-grained and relation-sensitive queries (*e.g.*, *Find a cabinet next to the TV with a dried potted plant on top*"). For dynamic trials, an LLM is given the scene layout and selects diverse targets and plausible relocation positions following common household habits.

Table 8: Real-world navigation. System performance under static and dynamic household settings.

Method	Static		Dynamic		
	SR (%) ↑	SPL (%) ↑	SR (%) ↑	SPL (%) ↑	Attempts ↓
DualMap [9]	63.3	58.2	56.7	18.1	2.1
Ours	**86.7**	**76.3**	**76.7**	**27.6**	**1.4**

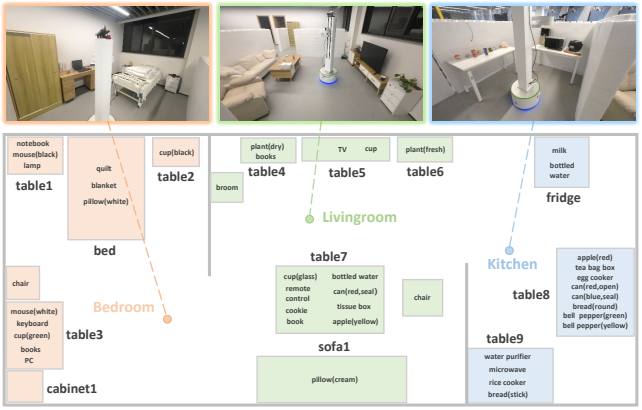


Figure 4: Real-world household environment. The three-room scene contains 13 semantic anchors and visually similar objects with diverse colors, shapes, and states.

As shown in Table 8, OASIS outperforms DualMap in success rate and SPL while requiring fewer relocation attempts. Across 60 trials, we observe four static and seven dynamic failures. Static failures include two VLM attribute confusions (*e.g.*, beige versus white pillows) and two mapping misses of small items (*e.g.*, a remote control). Of the dynamic failures, four involve similar visual or mapping errors, two arise when distant-room searches are restricted to local anchors, and one results from a missed transient detection. Overall, perception and long-range reasoning remain the primary failure modes.

5 Conclusion

We introduced OASIS, an online anchor-centric system for open-vocabulary retrieval and navigation in dynamic environments. Its multimodal memory and hierarchical LLM–VLM pipeline balance semantic expressiveness with online efficiency, while transient navigation observations support moved-object relocalization. Simulation and real-world experiments demonstrate improvements in retrieval accuracy, navigation success, and relocalization robustness over baselines, establishing anchor-centric memory as a practical foundation for long-horizon embodied intelligence.

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